

# Clément Soubrier

Postdoctoral Fellow in Mathematics

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## RESEARCH INTERESTS

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Mathematical biology, optimal transport and shape analysis.

## ACADEMIC POSITIONS

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2026–present      **Postdoctoral Fellow**, University of British Columbia, Vancouver.  
Supervisor: Khanh Dao Duc. Optimal transport methods for super-resolution biological imaging and high-dimensional genomic data.

## EDUCATION

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2022–2026      **PhD in Mathematics**, University of British Columbia, Vancouver.  
Thesis: *Mathematical modeling and analysis of biological shape heterogeneity and cellular growth*. Supervisor: Khanh Dao Duc.

2018–2022      **MSc, Engineering curriculum, Mathematics**, CentraleSupélec, Université Paris-Saclay.  
Maths–Physics and Research track.

2016–2018      **Classe préparatoire (MP)**, Lycée Louis-le-Grand, Paris.

2016      **Baccalauréat and Abitur (AbiBac)**, Lycée Richelieu, Rueil-Malmaison.

## PUBLICATIONS AND PREPRINTS

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2026      **C. Soubrier**, G. Woollard, A. Warren, K. Dao Duc. *Beyond Procrustes distances: a multilinear Gromov–Wasserstein distance capturing chirality*. Preprint, arXiv:2608.27774.

2025      E. Foxall, **C. Soubrier**. *Time of appearance of a large gap in a dynamic Poisson point process*. Preprint, arXiv:2512.04218.

2025      **C. Soubrier**, A. Madzvamuse, H. A. Eskandarian, K. Dao Duc. *Quantifying the organization and dynamics of *M. smegmatis* morphology from long-term time-lapse atomic force microscopy*. Preprint, bioRxiv.

2024      **C. Soubrier**, E. Foxall, L. Ciandrini, K. Dao Duc. *Optimal control of ribosome population for gene expression under periodic nutrient intake*. Journal of the Royal Society Interface, doi:10.1098/rsif.2023.0652.

## TALKS

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May 2026      *Integrating shape dynamics to enhance cell motility analysis*.  
BIRS workshop A Roadmap towards Developing Mechanobiochemical Models for Single and Collective Cell Migration through Complex Non-Isotropic Environments, Banff.

Mar. 2026      *The multilinear Gromov–Wasserstein distance and its application to chiral shape comparison*.  
Kantorovich Initiative Retreat, UBC, Vancouver.

July 2025      *Modeling the meiotic spindle using a spatial birth-death process*.  
SMB Annual Meeting, minisymposium Probability and stochastic processes in biology, Edmonton.

Apr. 2024      *Analyzing morphological properties of *M. smegmatis* using temporal data*.  
IAM Annual Retreat, UBC, Vancouver.

Sep. 2023      *End to end pipeline for cell shape dynamics analysis from atomic force microscopy time series*.  
BIRS workshop Mathematical Methods for Exploring and Analyzing Morphological Shapes across Biological Scales, Banff.

Jan. 2023      *Impact of ribosome population control on the dynamics of protein expression*.  
Joint Mathematics Meetings, Boston.

## TEACHING

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UBC: MATH 100 (2026), MATH 180 (2024,2025)

Small class  
instructor

UBC: MATH 406 (2024), MATH 307 (2023), MATH 152 (2023), MATH 253 (2022)

Teaching  
assistant

## AWARDS

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2022–2026 Four year Fellowship, UBC

2021 Globalink Research Internship Award, Mitacs

## LANGUAGES

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- French (native)
- English (fluent)
- German (fluent)